

FlowPilot improves design assurance across model scales

Matched architecture comparison using identical protocols, inventories, model identity and scoring within each pair.

Qwen3.6-27B

0.674 → 0.958

architecture uplift +0.284

GPT-5.4

0.742 → 0.953

architecture uplift +0.212

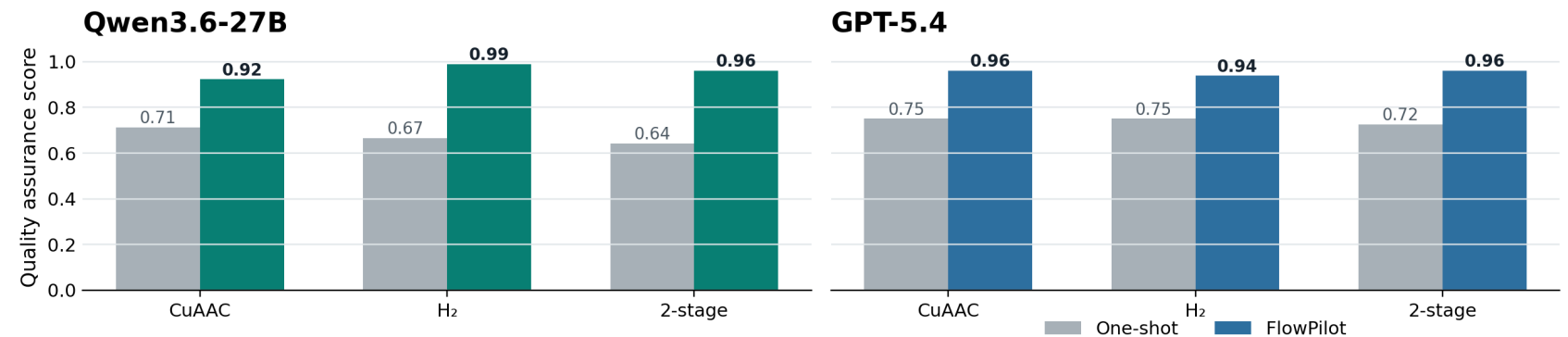
6 / 6

paired QA
improvements

All six full-pipeline outputs closed as
executable SCREEN designs.

The architecture improves assurance for every model-case pair

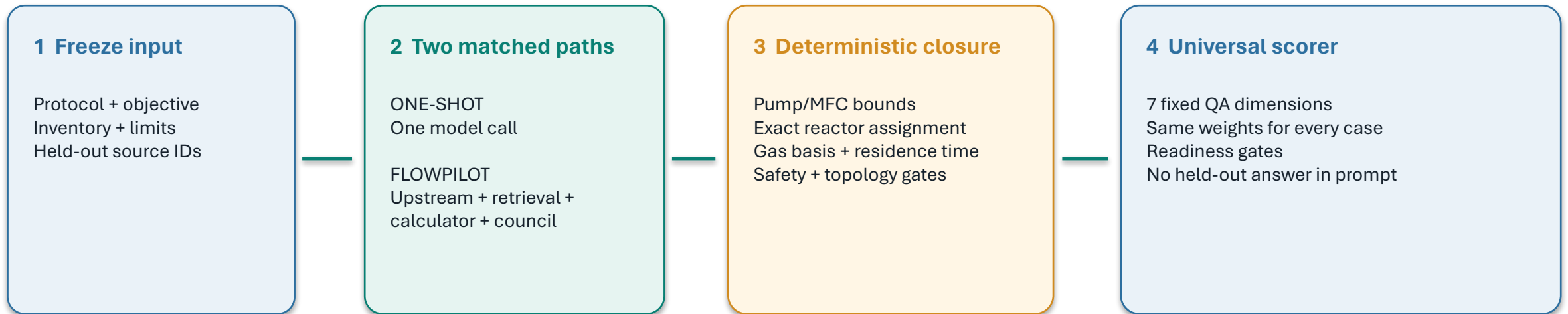
Same protocol, inventory, model, candidate budget and universal scorer within each pair



Interpretation The same model becomes more dependable when coupled to retrieval, engineering calculations, inventory enforcement, specialist review and a final deterministic contract.

How the matched benchmark was executed

The model is controlled within each pair; only the architecture changes.



Quality-assurance score



Hard gates cap readiness when schema, gas bookkeeping, geometry, pumps/tubing, topology, or safety fail.

Controlled variables: candidate budget = 1 | temperature = 0 where provider-supported | lexical retrieval | source answer excluded | same case-specific inventory

Why the architecture changes the outcome

One-shot numeric closure can be strong; FlowPilot adds traceable decision and process assurance.

FlowPilot's largest gains come from decision assurance and provenance



Observed interventions

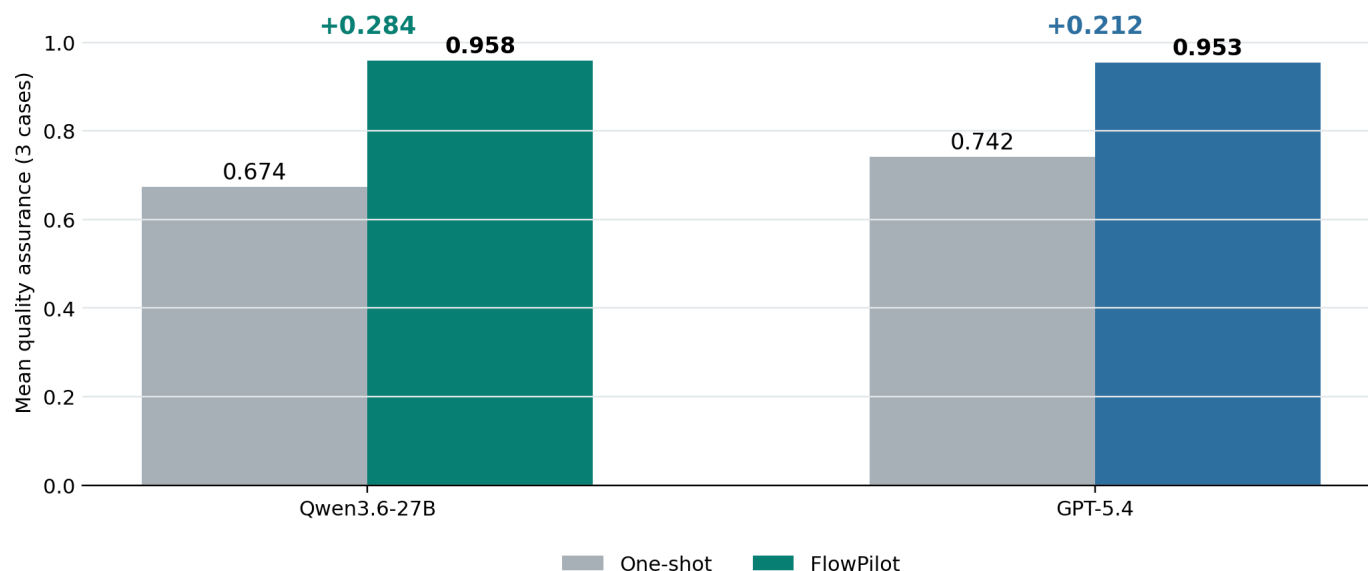
- Rejected unsupported residence-time reductions.
- Recomputed V/Q and stage-specific cumulative flows.
- Converted H₂ inlet/STP flow to in-channel flow and equivalents.
- Bound pumps, MFC, reactors, BPR and mixers to inventory IDs.
- Published topology only after final deterministic closure.

Result: 6/6 executable SCREEN contracts

Presentation conclusion: architecture, not model size alone

The local 27B model in FlowPilot reaches essentially the same mean assurance as GPT-5.4 in FlowPilot.

Architecture uplift persists from a 27B local model to GPT-5.4



Defensible claim

Across these three inventory-constrained cases, FlowPilot improved universal design-assurance scores for both Qwen3.6-27B and GPT-5.4, and produced executable screening contracts in every full-pipeline run.

Do not overclaim

- n = 3 cases and one run per cell
- no statistical significance claim
- scores assess design assurance, not reaction yield
- final designs remain SCREEN until wet-lab validation